

Ayushman Lohani

GenAI Engineer | Agentic RAG & LLM Systems | PyTorch, LangChain, FastAPI

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EDUCATION

University of Lucknow

Bachelor of Technology in Computer Science (Artificial Intelligence)

Lucknow, UP

Oct. 2023 – Sep 2027

EXPERIENCE

AI & Machine Learning Intern

July 2025 – Sep. 2025

National Institute of Electronics & Information Technology (NIELIT)

Online

- Built sequence-to-sequence Transformer from scratch in PyTorch (Multi-Head Attention, positional encoding) to replicate core GenAI architecture.
- Optimized training on 175k+ points with dynamic LR schedulers, reducing validation loss to 0.84 and convergence time by 25%.
- Refactored notebooks into modular production Python, enabling real-time inference and clean deployment handoff.

Machine Learning & Data Science Intern

Jul. 2025 – Aug. 2025

Info Bharat Interns

Online

- Engineered XGBoost forecasting pipeline on 17k+ temporal records, achieving R^2 0.96 for demand prediction.
- Standardized preprocessing (scaling, temporal alignment) to eliminate inconsistencies and improve reliability.

PROJECTS

Unitwise — Agentic RAG Study Assistant

[\[GitHub\]](#) [\[Live\]](#)

Python, LangChain, FastAPI, ChromaDB, Groq API, GPT OSS 120B, all-MiniLM-L6-v2

- Built agentic RAG pipeline with LangChain + ChromaDB (all-MiniLM-L6-v2 384-d, top-15) + Groq LLM, streaming cited answers with page numbers in <5s via SSE.
- Engineered ingestion: PyMuPDF → 1000/200 chunking → embeddings → SQLite; 3-tier syllabus gate + Groq key pool (25 req/min/key) blocks off-topic queries.
- Deployed FastAPI on HF Spaces with IP rate limiting (10/min, 3/10s via slowapi) + React 19 on Vercel with Supabase auth, serving 50+ users and 1000+ queries.

MCP Server — Multi-Tool LLM Backend

[\[GitHub\]](#)

Python, MCP SDK, MongoDB, Tavily API, DDGS, Scraping, Trafilatura, Pytest

- Built MCP server exposing 8 tools over streamable-HTTP (127.0.0.1:8000) - Tavily→DDGS search, trafilatura fetch, and MongoDB notes CRUD for LLM agents.
- Implemented 2-tier fallbacks (plain→stealth browser, Tavily→DDGS) handling 403/429/503 and Cloudflare challenges, with 15k-char cap and 4 mocked pytest suites.

RBI Sentinel — Market Volatility Forecaster

[\[GitHub\]](#) [\[Live\]](#)

Python, FinBERT, Scikit-Learn, Pandas, Statsmodels, Streamlit

- Engineered FinBERT pipeline with 510-token sliding window to score 140 RBI speeches/MPC minutes (2020–2026), using |score| intensity as signal.
- Trained Random Forest (300 trees, depth 4, balanced) on 5 features (intensity, VIX lag1, Nifty 5d return/std, Is_MPC) forward-merged via `merge_asof`, R^2 0.91 on 2025 hold-out.
- Deployed Streamlit dashboard for live inference on pasted RBI text with Yahoo Finance VIX/Nifty context.

TECHNICAL SKILLS

Languages: Python, SQL, C++

ML/AI Frameworks: PyTorch, Scikit-Learn, LangChain, LangGraph, HuggingFace, XGBoost, FinBERT

Concepts: Large Language Models, RAG, NLP, Agentic AI, Embeddings, Vector Databases, Prompt Engineering

Libraries, Tools: Pandas, NumPy, FastAPI, Streamlit, ChromaDB, MCP SDK, MongoDB, Git

CERTIFICATIONS

Agentic AI Certified Foundations Associate — **Oracle**

Artificial Intelligence Fundamentals — **IBM**

Certified Data Science Professional — **Oracle**